MYSQL PRACTICAL EXAM

Company Employee Database

Student Name: _____ Date: _____

Batch: _____ Time: 90 minutes

Total Marks: 40 Attempt all questions. Write SQL queries only; answers are not provided.

Part 1 — Create Database and Table

Create database **CompanyDB** and table **Employees** with columns: employee_id INT PRIMARY KEY, employee_name VARCHAR(50), department VARCHAR(30), job_title VARCHAR(40), city VARCHAR(30), salary INT, experience_years INT, joining_year INT.

Insert these 20 records:

ID	Employee Name	Department	Job Title	City	Salary	Exp.	Year
101	Aarav Shah	IT	Developer	Ahmedabad	65000	3	2022
102	Diya Patel	HR	HR Executive	Rajkot	42000	4	2021
103	Rohan Mehta	Sales	Sales Executive	Surat	38000	2	2024
104	Mira Desai	Finance	Accountant	Vadodara	52000	5	2020
105	Kabir Joshi	IT	Tester	Rajkot	48000	3	2023
106	Anaya Trivedi	Marketing	Marketing Executive	Ahmedabad	45000	2	2024
107	Vivaan Shah	Sales	Sales Manager	Surat	72000	7	2019
108	Isha Parmar	IT	Developer	Vadodara	68000	4	2021
109	Arjun Vyas	HR	Recruiter	Ahmedabad	40000	2	2023
110	Sara Bhatt	Finance	Financial Analyst	Surat	61000	6	2020
111	Krish Patel	IT	Support Engineer	Rajkot	39000	1	2025
112	Myra Shah	Marketing	Marketing Manager	Vadodara	70000	6	2019
113	Dev Modi	Sales	Sales Executive	Ahmedabad	41000	3	2022
114	Aadhya Shah	Finance	Accountant	Rajkot	50000	4	2021
115	Reyansh Dave	IT	Developer	Surat	75000	5	2020
116	Kiara Mehta	HR	HR Manager	Vadodara	66000	8	2018
117	Dhruv Patel	Sales	Sales Executive	Rajkot	36000	1	2025
118	Pari Joshi	Marketing	Content Specialist	Surat	44000	3	2022
119	Neil Desai	Finance	Finance Manager	Ahmedabad	80000	9	2017
120	Tara Shah	IT	Developer	Vadodara	70000	4	2022

Part 2 — SQL Query Questions

Each question carries 1 mark. Total: 40 marks.

A. SELECT and WHERE

- Display all employee records.
- Display employee_name, department and salary only.
- Find all employees in the IT department.
- Find employees with salary greater than 60,000.
- Find employees with at least 3 years of experience.
- Find employees working in Rajkot.
- Find employees who joined in 2022 or later.
- Find employees whose salary is between 40,000 and 65,000.

B. AND, OR, NOT

- Find IT employees whose salary is above 60,000.
- Find employees from Rajkot or Surat.
- Find employees who are not in Sales.
- Find employees with at least 4 years' experience and salary above 50,000.
- Find HR or Finance employees earning at least 50,000.
- Find employees not from Ahmedabad.

C. ORDER BY and LIMIT

- Sort employees by salary from highest to lowest.
- Sort employees by name alphabetically.
- Display the 5 highest-paid employees.
- Display the 3 lowest-paid employees.
- Display the 5 most recently joined employees.

D. Aggregate Functions

- Count all employees.
- Find the highest salary.
- Find the lowest salary.
- Calculate average salary.
- Calculate the total of all salaries.
- Find average experience.

E. GROUP BY

- Count employees in each department.
- Find average salary for each department.
- Find maximum salary in each department.
- Find minimum salary in each city.
- Calculate total salary for each department.
- Count employees in each city.

F. HAVING

- Show departments with more than 3 employees.
- Show departments whose average salary exceeds 55,000.

- Show cities with at least 4 employees.
- Show departments whose total salary exceeds 150,000.

G. Combined Queries

- Show the 3 departments with the highest average salary.
- Show department-wise employee count, highest count first.
- Show cities with average salary above 50,000, highest average first.
- Find departments with more than 2 employees and average salary above 45,000.
- Display the 5 highest-paid IT employees.
- Find employees from Rajkot or Surat earning above 45,000, sorted by salary descending.
- Show department-wise count and average salary only for departments averaging above 45,000.
- Show departments whose minimum salary exceeds 40,000, sorted by minimum salary ascending.